

Language-conditioned Spatial Affordance Refinement for Diffusion-based Grasp Detection

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<https://github.com/TBLboy/language-driven-grasp-detection>

Abstract

Language-driven grasp detection requires grounding a natural-language grasping instruction to a manipulable region in an RGB image. We build on the official LGDM baseline and repair two implementation gaps: the ALBEF text condition is not connected to the grasp decoder, and the diffusion loss is not back-propagated. We then propose LSAR, a lightweight spatial affordance refinement module that injects a small residual correction into the decoder bottleneck before grasp generation. On a scene-disjoint 10,000-scene subset of Grasp-Anything++, LSAR improves repeated-evaluation correct grasps from 470.0 to 678.0 and 661.7 under two training seeds, while preserving the official grasp representation and evaluation protocol.

1. Introduction

Language-driven grasp detection requires predicting a five-parameter grasp rectangle

$$G = (x, y, w, h, \theta) \quad (1)$$

for the object or part selected by a free-form instruction. We start from the diffusion-based LGDM formulation. In its released code, the vision-language fusion output is computed but not injected into the grasp decoder, and the diffusion loss is not optimized. We define a clean baseline that restores the diffusion objective and introduce LSAR, a compact residual module that makes the language condition spatially explicit. The main contribution is the LSAR conditioning mechanism; the repaired baseline, repeated evaluations, and ablation are provided to make the effect reproducible and interpretable.

2. Related Work

Grasp detection methods based on dense quality, angle, and width maps [1, 3] provide the representation used by many language-conditioned systems.

Language-driven Grasp Detection (LGD) [4] introduces Grasp-Anything++ [5] and a diffusion-based conditional model. Vision-language encoders such as ALBEF [2] align semantic concepts but do not directly encode graspable regions. Our module adapts the language-conditioned spatial branch without changing the pretrained encoders or the generative head.

3. Method

3.1. Problem Setup

An input sample is (I, T) , where I is an RGB image and T is a grasping instruction. Ground-truth rectangles are converted into dense maps pos , $\cos(2\theta)$, $\sin(2\theta)$, and normalized width; post-processing decodes the predicted dense maps into a rectangle.

3.2. Clean LGDM Baseline

LGDM computes $y_{\text{view}} \in \mathbb{R}^{8 \times 19 \times 19}$ with ALBEF, but the released decoder does not use it, and the upstream training script does not back-propagate the diffusion loss. Our baseline keeps the official decoder structure and optimizes

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{diffusion}} + \mathcal{L}_{\text{cos}} + \mathcal{L}_{\text{sin}} + \mathcal{L}_{\text{width}}. \quad (2)$$

We also re-enable a raw “plain- y ” injection into the GG-CNN decoder feature F_v for the language-conditioning ablation.

3.3. LSAR

Semantic alignment does not identify which part of a target affords a grasp. LSAR combines the decoder feature with the language map and learns a small residual:

$$Z = \text{concat}(F_v, y_{\text{view}}), \quad (3)$$

$$H = \text{ReLU}(\text{conv}_{3 \times 3}(\text{GN}(\text{ReLU}(\text{conv}_{3 \times 3}(Z))))), \quad (4)$$

$$\Delta = s \text{conv}_{1 \times 1}(H), \quad (5)$$

$$F' = \text{ReLU}(F_v + \Delta). \quad (6)$$

We fix $s = 0.01$ to keep the residual perturbation small. The module also emits a 19×19 affordance map $A = \text{conv}_{1 \times 1}(H)$ for an auxiliary term studied below.

